KEERTHANA THUMMALA

Software Engineer | Full-Stack Developer | AI/ML | Data Analysis

k.thummala08@gmail.com | +91 9581944004 | LinkedIn | GitHub | Portfolio

SUMMARY

Software Engineering student at The University of Queensland (expected 2027) with experience in full-stack development, AI/RAG applications, cloud-native deployment, and data analysis. Builds end-to-end, from React front ends and FastAPI backends to LangGraph agents and retrieval pipelines, deployed as containerised microservices on Kubernetes (GKE). Contributed to a 20% reduction in infrastructure costs and a 25% improvement in engineering efficiency through CI/CD work at RCS Tech.

EDUCATION

Bachelor of Science in Software Engineering | The University of Queensland | 2023 - 2027 | GPA: 5.3/7.0

Relevant coursework: Full-Stack Development, Data Structures & Algorithms, Agile Methodologies, Machine Learning, Data Science, LLMs.

TECHNICAL SKILLS

- **Languages:** Python, Java (OOP), JavaScript, TypeScript, SQL
- **Frontend & Backend:** React.js, Tailwind CSS, HTML5/CSS3, Node.js, FastAPI, RESTful APIs
- **Databases & Caching:** MySQL, MongoDB, ChromaDB (Vector DB), Redis
- **AI / ML:** LangGraph, LangChain, RAG, Gemini API, Prompt Engineering, Ollama, Scikit-learn, Pandas, NumPy, PCA
- **Data Analysis:** Exploratory Data Analysis, Feature Engineering, Matplotlib/Seaborn, Model Evaluation
- **AI-Assisted Development:** Claude Code, GitHub Copilot, Gemini, ChatGPT, Lovable, Canva, DeepSeek, LLM API integration, prompt engineering for coding workflows
- **Tools & Methodologies:** Docker, Kubernetes (GKE), CI/CD, Git/GitHub, Agile/Scrum, Jira, SDLC, System Design, Microservices

WORK EXPERIENCE

Full Stack Developer | RCS Tech LLP | 12/25 - 04/26

- Engineered scalable web architecture for the TRAXX platform, reducing infrastructure expenses by 20%.
- Streamlined development workflows and CI/CD pipelines, improving team engineering efficiency by 25%.
- Led sprint planning and code reviews in an Agile framework to keep delivery aligned with client specifications.

Software Development Intern | CodSoft | 01/25 - 02/25

- Built modular applications in Java and Python, improving code reusability and system scalability.
- Refined application logic and memory management, increasing execution efficiency by 25%.

MAJOR PROJECTS

OrbitOps Assistant | *Python, FastAPI, React, LangGraph, ChromaDB, MySQL, Redis, Docker, Kubernetes (GKE), Gemini API*

- Built an AI operations assistant that answers questions from runbooks with cited sources (hybrid BM25 + vector RAG), researches GitHub and HackerNews through a LangGraph agent, and diagnoses its own Kubernetes cluster with a one-click health check.
- Re-architected a single-process app into 4 containerised microservices (frontend, chat, auth, ops) on Google Kubernetes Engine, backed by MySQL, ChromaDB and Redis on persistent storage.
- Configured horizontal pod and cluster autoscaling (2→5 pods, 2→4 nodes), health probes and disruption budgets; verified zero failed requests across a rolling update and rollback, and recovery from a node failure.
- Secured the platform with hashed-password accounts, an admin audit page, least-privilege RBAC, Kubernetes Secrets and prompt-injection guardrails, with automatic fallback to a second LLM when the primary is overloaded.

Nutrient Profile Classification — Data Analysis | *Python, Pandas, Scikit-learn, Matplotlib/Seaborn, PCA*

- Cleaned a messy 1,590-item, 272-column nutrition dataset into an analysis-ready format.
- Engineered features (protein-per-calorie, sodium-to-potassium ratio) and built KNN and Random Forest classifiers.
- Identified a data leakage issue inflating model accuracy and proposed a corrected feature set.

Resume Tracker Extension | *React.js, Java, JSON, HTML/CSS*

- Built a browser extension with secure user registration/login and a dynamic job-application tracking system, using AI-assisted coding tools to accelerate development and debugging.
- Used JSON to persist and reload user data seamlessly across sessions.

RESEARCH

Honours Thesis: Benchmarking Hardness in Package-to-Truck Allocation for Near-Term Quantum Optimisation

Supervisor: Prof. Eleanor Rieffel | The University of Queensland | 2026

- Designed a benchmark suite of package-to-truck allocation problems (100–10,000 packages) grounded in real Australian data, with a hardness metric validated against solver statistics; ranked instances via OR-Tools CP-SAT, scoping future quantum work (QAOA) on the hardest set.

Tools: Python, OR-Tools, Qiskit, python-docx